Effect of BGP Local Preference on a Route Between AS200 and AS30

Otto Karppinen
TIES322

***Abstract*—This paper looks at how the Border Gateway Protocol local preference attribute affects route selection in a network. The test was run at the Kathara Small Internet Lab. In the setup the default route that came directly from AS30 had a local preference of 90 and AS200 picked the alternate route through AS20. The local preference was then increased to 200. After that change the next hop switched from AS20 to AS30 and the traceroute from AS200 to a host in AS30 went from four entries, to three. The result shows how a local routing policy can change the forwarding path without altering the topology.**

I. Introduction

BGP is used to share routing information between autonomous systems [1]. BGP does not always select the physically shortest route. It uses different route attributes and routing policies when selecting a route [2]. Because of this, an available direct connection may not always be used.

In this experiment, I studied the route between AS200 and AS30. The purpose was to test if changing the local preference value could make the route shorter. I expected that a higher local preference for the direct AS30 route would remove AS20 from the traceroute path.

II. Background

The LOCAL_PREF attribute shows the preference given to a route within an AS. A route with a higher value is chosen over a route with a lower value during the BGP decision process [1]. This helps a network operator decide for example which upstream provider should handle traffic. These policy choices can also affect the stability of interdomain routing [4], [5]. In operational networks, route filtering and clear routing policies are important for both reliability and security [3].

Traceroute was used to see the path that data takes at the IP level. The result, from Traceroute shows which routers reply to the probes but Traceroute alone does not explain why BGP chose a route. Therefore the results from Traceroute were examined with the BGP table.

III. Experimental Setup

The experiment used the Kathará Small Internet Lab. The source was the `client` host at `200.2.0.2` in AS200, and the destination was `ldns` at `30.3.3.2` in AS30. AS200 had BGP connections to both AS20 and AS30.

In the original configuration, the default route received from AS30 through next hop `11.0.0.10` had a local preference of 90. The route through AS20, with next hop `11.0.0.34`, was selected. The only experimental change was replacing `set local-preference 90` with `set local-preference 200` in the AS200 BGP configuration. The lab was restarted after the change. The BGP table, routing decision, ping result, and traceroute were recorded in both configurations.

IV. Results

Table I sums up the measurements. Before the modification, the BGP table marked `11.0.0.34` as the best next hop. Traceroute then passed through AS200, AS20, and AS30 before reaching the destination. After the local preference was increased to 200, BGP selected the direct AS30 next hop `11.0.0.10`. The address belonging to AS20 disappeared from the traceroute.

The original traceroute was `200.2.0.1`, `11.0.0.34`, `11.0.0.10`, and `30.3.3.2`. After the configuration change, the traceroute was `200.2.0.1`, `11.0.0.10`, and `30.3.3.2`.

TABLE I
Route before and after the BGP policy change. LOCAL_PREF is the value configured for the route received from AS30; the route from AS20 used the default value of 100.

LOCAL_PREF	Next hop	AS path	Hops
90	11.0.0.34	200–20–30	4
200	11.0.0.10	200–30	3

The number of traceroute entries decreased from four to three, corresponding to one fewer intermediate router. Both ping tests delivered four out of four packets, so packet loss was zero in both configurations. The measured average round-trip time changed from 1.688 ms to 1.470 ms, but the small number of samples does not support a general performance conclusion.

V. Discussion

The result was what I expected. The direct route to AS30 was available in the original configuration, but its local preference was only 90. Because of this, AS200 selected the route through AS20. When I changed the value to 200, AS200 selected the direct route to AS30. The new traceroute was one hop shorter. This shows that BGP does not always select the physically shortest route. The configured routing policy also affects the selection.

This experiment was small. It used an emulated network, one source and destination, and only one configuration change. Traceroute also shows only the routers that answer its packets.

More destinations and more measurements could give better information about network delay and reliability. However the two measurements were enough to show how local preference changed the selected route.

VI. Conclusion

Raising the local preference of the direct AS30 route from 90 to 200 caused AS200 to stop forwarding the tested traffic through AS20. The traceroute was shortened by one intermediate router while connectivity remained intact. The experiment provides a practical example of how a single BGP policy attribute can alter the path used by network traffic.

References

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